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Co-optimization of Coincident Peak Management and Ancillary Services for Flexible Data Centers

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***Abstract*—Flexible Data Centers (FDCs) have become an important demand-side resource in power systems as their scale and flexibility continue to grow. By rescheduling interruptible and delayable computing tasks, FDCs can rapidly adjust power consumption and participate in electricity market activities such as demand response and ancillary services (AS). Meanwhile, many electricity markets adopt a coincident peak (CP) mechanism, under which FDCs may incur substantial charges if they fail to reduce load during peak hours. As a result, FDC operations must balance computing performance, market revenue, and CP risk. Under these conditions, the major challenge is to determine operating strategies that can simultaneously satisfy computing needs, support market participation, and reduce peak-related penalties under uncertain CP events and multiple market objectives. This paper proposes a unified co-optimization methodology for FDC operations. A high-fidelity physical–economic model is first constructed to capture the relationship between input power, computational throughput, and economic returns. Then, a model-agnostic three-stage CP prediction strategy is developed to identify periods with high CP risk. Building on these components, a risk-driven and prediction-aware co-optimization methodology (RPCM) is developed to jointly optimize baseline workloads, AS bids, and peak-avoidance actions. Case studies in the ERCOT market in Texas show that the proposed method achieves significantly higher overall profit than traditional approaches that treat these decisions separately, as it can effectively avoid CP charges while increasing AS revenue. The results demonstrate that the proposed methodology provides a practical and robust decision tool for FDC operators under multi-market conditions.**

***Index Terms*—Flexible data centers, coincident peak, ancillary services, two-stage stochastic programming**

NOMENCLATURE

Indices

ω	Scenario index representing the CP hour.
h	Hour index in the conditional peak-hour probability vector $P_{h\|\mathrm{day}}$.
i	Index of computing unit.
k	Index of AS product.
s	Index of operating mode of a computing unit (1:Off, 2:Low, 3:Normal, 4:High).
t	Time index (hour of day).

Sets

Ω	Set of peak-hour scenarios within a day.
AS_D	Set of downward-regulation ASs.
AS_U	Set of upward-regulation ASs.
D_{pred}	Feature set for the CP prediction models.
I	Set of computing units.
K	Set of AS products.
S	Set of operating modes.
T	Set of hourly time periods in the scheduling horizon.

Physical Layer Parameters

α	Activity factor of the chip.
η	Efficiency of the power converter.
κ	Equivalent thermal slope relating grid-side input power to junction temperature rise under the aggregated linear approximation.
κ_{chip}	Equivalent thermal coefficient relating chip-side dissipated power to junction temperature rise.
A_s	Slope coefficient of the piecewise linear $H(P)$ segment in mode s.
a_T	Temperature coefficient of leakage power.
B_s	Intercept coefficient of the piecewise linear $H(P)$ segment in mode s.
C_{eff}	Equivalent switching capacitance.
f	Chip operating frequency.
H	Average effective computational throughput rate after overhead discount.
$H(P)$	Power–throughput mapping function from input power to effective throughput.
$H_{\max}$	Maximum throughput in a given operating mode.
$H_{\min}$	Minimum throughput in a given operating mode.
H_{base}	Average raw computational throughput rate before platform overhead.
k_T	Slope of fan power with respect to junction temperature.
k_{thru}	Throughput coefficient per clock cycle (tasks per cycle).
L_s	Minimum feasible power of a single unit in operating mode s.
P	Grid-side input power of the data center.
$P_{\max}$	Maximum device power in a given operating mode.
$P_{\min}$	Minimum device power in a given operating mode.
P_{aux}	Auxiliary overhead power.
P_{chip}	Total chip power consumption.
P_{dc}	DC output power of the power converter.
P_{dyn}	Dynamic power consumption due to active computation.
$P_{fan,0}$	Baseline fan power at nominal temperature.
P_{fan}	Fan power consumption.
$P_{leak,0}$	Baseline leakage power at nominal temperature T_{nom}.
P_{leak}	Static leakage power consumption.
p_{pool}	Non-computational overhead ratio on the platform side.
T_j	Junction temperature of the chip.
T_{amb}	Ambient air temperature.
T_{nom}	Nominal reference temperature.
U_s	Maximum feasible power of a single unit in operating mode s.
V_{nom}	Nominal motherboard supply voltage.

Economic and Market Parameters

$\alpha_{k,t}$	Capacity price of AS product k at hour t.
Δt	Length of the scheduling interval.
Π_{DA}	Day-ahead profit component.
$\Pi_{RT}(\omega)$	Real-time profit component under scenario ω.
θ	Risk tolerance threshold.
C^{CP}	CP capacity penalty rate.
D	Network mining difficulty.

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$E[F]$	Expected transaction-fee reward per block.
f_{pool}	Mining-pool fee rate.
M	Revenue rate associated with effective throughput.
m	Revenue coefficient per unit effective throughput rate per unit time.
M_{day}	Trained peak-day probability prediction model.
M_{hour}	Trained peak-hour probability prediction model.
p_t^{DA}	Day-ahead electricity price at hour t.
p_t^{RT}	Real-time electricity price at hour t.
P_{asset}	Market price of the target digital asset.
R^{blk}	Fixed protocol block reward.
s_l	Invalid-share loss rate on the operator side.
$W_{t,\omega}$	Scenario-dependent binary CP indicator parameter: 1 if hour t is the CP settlement hour in scenario ω, else 0.
Decision Variables	
dP_t^{ω}	Aggregated AS down-regulation power at hour t under scenario ω.
$dw_{i,t,s}^{\omega}$	AS down-regulation deployment on unit i, mode s, hour t, scenario ω.
gP_t^{ω}	Aggregated CP-avoidance curtailment power at hour t under scenario ω.
$gw_{i,t,s}^{\omega}$	Additional CP-avoidance power curtailment on unit i, mode s, hour t, scenario ω.
H_t^{base}	Auxiliary quantity representing the baseline total average computational throughput rate during interval t, uniquely determined by the baseline power decisions.
$H_{t,\omega}^{eff}$	Auxiliary quantity representing the real-time effective total computational throughput rate during interval t under scenario ω, uniquely determined by the real-time power adjustments.
P_t^{base}	Baseline total power consumption of the data center at hour t.
$P_{\max,t}$	Maximum accessible total power at hour t implied by chosen modes.
$P_{\min,t}$	Minimum accessible total power at hour t implied by chosen modes.
$p_{i,t,s}$	Baseline operating power of unit i in mode s at hour t.
$P_{t,\omega}^{eff}$	Real-time effective total power at hour t under scenario ω.
Q_t^{DA}	Day-ahead energy purchase quantity at hour t.
$r_{k,t}$	Offered AS capacity of type k at hour t.
uP_t^{ω}	Aggregated AS up-regulation power at hour t under scenario ω.
$uw_{i,t,s}^{\omega}$	AS up-regulation deployment on unit i, mode s, hour t, scenario ω.
$y_{i,t,s}$	Binary mode-selection variable: 1 if unit i operates in mode s at hour t, else 0.
Random Variables	
π_{ω}	Conditional probability of scenario ω.
F	Random transaction-fee reward per block.
P_{day}	Probability that the target day is a CP day (peak-day probability).
$P_{h\|\text{day}}$	Conditional probability that hour h is the CP hour, given a CP day.

I. Introduction

AS data centers consume an increasingly significant share of global electricity, their operational behavior has transitioned from being passive loads to becoming pivotal factors influencing grid operational efficiency and stability [1]. The power fluctuations of data centers can directly impact system peak-to-valley spreads, reserve capacity requirements, and electricity price trends. As a result, optimizing the way data centers interact with the grid is of strategic importance. It offers potential benefits such as improved energy efficiency, enhanced integration of renewable energy sources, and stronger support for the sustainable development of the digital economy.

Traditionally, data centers have been regarded as inflexible consumers with limited controllability [2]. However, advances in computing technologies and rising electricity costs have driven many data centers to adopt more flexible operating strategies [3]. By dynamically scheduling interruptible or delayable computing tasks [4], these facilities can respond to electricity price or grid dispatch signals on a minute-to-hour timescale, effectively achieving peak shaving during high-price periods and increasing utilization during off-peak hours. Typical examples include High-Performance Computing (HPC) centers in drug discovery [5], rendering farms in the media industry [6], and price-sensitive cryptocurrency mining facilities [7].

This shift in the operational paradigm has led to the emergence of cost-aware Flexible Data Centers (FDCs), which aim to optimize both computing revenue and energy market participation. Prior studies have shown that FDCs can reduce electricity costs by approximately 10%–30% [7] while simultaneously contributing to grid stability through rapid power modulation. These dual benefits not only improve economic outcomes for operators but also strengthen the flexibility and reliability of the grid.

In the large-scale setting considered in this paper, FDCs are modeled as transmission- or sub-transmission-level flexible loads because of their high and controllable power demand. This demand-side flexibility opens up new economic opportunities for FDCs, but it also introduces operational complexities. Operators must balance between maintaining core computing performance and responding to various grid-side incentives. On the one hand, FDCs can offer valuable ancillary services (AS), including reserve capacity and fast frequency regulation, by leveraging their ability to quickly adjust power usage [7], [8]. On the other hand, a coincident peak (CP) mechanism is prevalent in many electricity markets worldwide [9], [10]. This mechanism requires users to pay high capacity charges for their electricity consumption during the several hours of highest system load [11], [12]. As a result, their operating decisions can affect system peak demand, reserve requirements, and market-based operating conditions. For large-scale flexible loads, balancing these two market opportunities is key to their market-based operation [13]. Therefore, how to jointly optimize core computing revenue and market participation within a unified methodology, while resolving the inherent tension between the capacity commitments required for AS and the imperative of CP avoidance, remains one of the most pressing challenges in the operation of FDCs [10].

To address this issue, the formulation of economic operation strategies for FDCs primarily faces three core challenges. Firstly, regarding physical layer modeling, current work analyzing the participation of large-scale FDCs in grid services typically abstracts them as aggregated units that can be entirely interrupted [7]. However, this black-box simplification obscures the critical operational realities of FDCs. Specifically, most existing models fail to capture the nonlinear conversion efficiency between input power and computational throughput. This lack of fidelity leads to distorted estimations of

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the marginal opportunity costs for providing AS, resulting in bidding strategies that are either physically infeasible or economically suboptimal.

Secondly, regarding the integration of prediction and optimization, the methods for incorporating probabilistic CP event predictions into decision models are still insufficient. Existing methods fall into two main categories. One is to embed the probabilities output by data-driven supervised learning [14]–[16] into deterministic optimization as heuristic weights [17]. However, this approach has limited practical applicability, as it fails to capture the complex, non-linear nature of CP mechanisms and is highly sensitive to probability calibration errors. The other category employs decision-integrated methods, such as approximate dynamic programming [18] or two-stage stochastic programming (2SSP) [10], [19]. Although such methods theoretically pursue optimality, they either demand rigorous prediction accuracy or treat scenario generation and optimization decisions as separate stages, leading to a lack of robustness in the overall decision-making. Therefore, despite stand-alone prediction models [20], a systematic, integrated prediction–decision strategy that is both robust and practical is still lacking.

Finally, regarding economic-layer decisions, existing studies mostly investigate strategies for FDC participation in a single market. Some cutting-edge studies focus on designing complex control strategies for FDCs to provide AS like frequency regulation while ensuring quality of service, but they do not consider CP risks [8]. Although research on cost-sensitive loads has recognized the importance of avoiding CP charges [21], [22], their models are limited to singularly avoiding CP penalties through means such as workload shifting on the computation side, local generation [21], or thermal pre-cooling on the facility side [22]. These studies analyze these objectives independently from AS opportunities, lacking a unified perspective for co-optimization. Consequently, a unified decision methodology that systematically studies the synergy and balances between these two economic drivers is still absent.

The overarching challenge, therefore, lies in effectively bridging the stochastic nature of sparse CP events with the non-linear physical constraints of FDCs. A methodology is notably absent to support operators in navigating the high-stakes balance between AS revenue and CP penalties. This paper aims to construct a unified decision methodology for the co-optimization of FDCs' participation in AS and CP management. To the best of our knowledge, this paper is among the first studies to systematically model and optimize the complex economic balances between CP management and AS for FDCs. The core contributions are as follows:

- A modular physical-economic model is proposed for FDC operation. It combines a high-fidelity physical layer that captures the power-to-throughput relationship with clear physical interpretability and a configurable economic layer that maps throughput into economic value. The economic layer is instantiated as a tractable linear throughput-to-revenue model for throughput-oriented flexible computing services. The resulting formulation provides physically interpretable marginal costs and operational constraints for market decision-making.
- A model-agnostic three-stage CP prediction strategy is designed to provide a decision-oriented probabilistic interface for FDC operation. The strategy employs rule-based filtering, peak-day identification, and peak-hour localization to transform heterogeneous prediction outputs into operationally usable risk quantities. In particular, the day-level probability is used to switch between the high-risk stochastic branch and the low/no-risk deterministic branch, while the conditional hourly probabilities are converted into scenario probabilities for the stochastic model. This design preserves modularity of the forecasting module while enabling structured prediction-to-decision coupling under CP uncertainty.
- A risk-driven and prediction-aware co-optimization methodology (RPCM) is developed, in which CP risk predictions are explicitly incorporated into the adaptive decision process to balance decision robustness with computational efficiency. Specifically, the model employs an adaptive mechanism that activates a stochastic formulation during high-risk periods and switches to a computationally efficient deterministic formulation when risk is low. This approach effectively resolves the inherent conflict between maintaining capacity commitments for AS and executing load curtailment for CP avoidance.

The remainder of this paper is organized as follows. Section II details the derivation of the physical-economic model. Section III provides an in-depth analysis of the CP mechanism and constructs a general CP event prediction strategy. Section IV establishes the co-optimization methodology and presents the mathematical formulation of the RPCM. Section V conducts a case study in the ERCOT market in Texas to validate the effectiveness of the proposed model and methodology. Section VI concludes the paper.

II. Modular Physical–Economic Model of FDC

This section develops the core physical and economic models of the FDC. It first develops a high-fidelity physical model to quantitatively characterize the P-H relationship. Subsequently, it builds a general economic model to convert H into M, providing the key physical and economic constraints for the co-optimization.

A. Physical Modeling

The adjustability of FDCs stems from the physical controllability of their equipment and the inherent delay tolerance of their services. Therefore, this section decomposes the total power into the consumption of each subsystem, using computing nodes as the basic units to build an end-to-end physical mapping from the grid-side input power P to the effective computational throughput H.

1) Energy Flow and Power Decomposition: Grid input power undergoes conversion losses and auxiliary consumption before reaching the chips. The initial AC-DC conversion is modeled as [23]:

$$P_{dc} = \eta P \tag{1}$$

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Subsequently, the net power delivered to the chips is determined by subtracting auxiliary and cooling overheads [24]:

$$P_{chip} = P_{dc} - P_{aux} - P_{fan} \tag{2}$$

2) Chip Power Consumption Decomposition: Chip power consumption can be divided into two parts [25]:

$$P_{chip} = P_{dyn} + P_{leak} \tag{3}$$

Since voltage regulation introduces only a small relative error in dynamic power consumption, we adopt the common engineering simplification of treating the motherboard supply voltage as a constant at V_{nom} [26]. This yields a linear relationship between dynamic power and the chip operating frequency [25]:

$$P_{dyn} = \alpha C_{eff} V_{nom}^2 f \tag{4}$$

Since leakage power is sensitive to temperature, it is first-order linearly approximated around the nominal temperature T_{nom} [27]:

$$P_{leak} \approx P_{leak,0} + a_T(T_j - T_{nom}) \tag{5}$$

3) Chassis Fan and Thermal Control: The power consumption of the cooling system is directly related to the chip's operating temperature and cooling efficiency. Its thermal control strategy is often described using a piecewise linear model in engineering [28]:

$$P_{fan} = P_{fan,0} + k_T \max(0, T_j - T_{nom}) \tag{6}$$

4) Effective Computational Throughput: Whether it is a CPU, GPU, or ASIC, under fixed voltage and architecture conditions, H_{base} and f can be approximated as a linear relationship [29]:

$$H_{base} = k_{thru} f \tag{7}$$

Considering orchestration, network, and protocol stack overheads, the effective throughput recorded on the platform side will have a certain discount [30]:

$$H = H_{base}(1 - p_{pool}) \tag{8}$$

5) Thermal Path Approximation and Junction Temperature: When the fan speed is fixed, the chip-to-ambient thermal path can be approximated by a linear equivalent model [31]. Thus,

$$T_j \approx T_{amb} + \kappa_{chip} P_{chip} \tag{9}$$

Under the fixed-mode approximation adopted here, P_{chip} can be expressed as an affine function of the grid-side input power through (1), (2), and (6), yielding the equivalent form:

$$T_j \approx T_{amb} + \kappa P \tag{10}$$

where κ is the corresponding equivalent thermal slope.

6) Power to Throughput Mapping Model: Assuming that the ambient temperature is stable at the nominal value, and by integrating the independent modeling of power flow, chip power consumption, cooling system, and the relationship between computational throughput and frequency described above, this section derives the linear affine relationship between P and H:

$$\begin{aligned} H(P) = {} & \frac{k_{thru}(1-p_{pool})}{\alpha C_{eff} V_{nom}^2} \left(\eta - (a_T + k_T)\kappa\right) P \\ & - \frac{k_{thru}(1-p_{pool})}{\alpha C_{eff} V_{nom}^2} \left(P_{aux} + P_{fan,0} + P_{leak,0}\right) \end{aligned} \tag{11}$$

In a fixed operating mode s, all parameters are constant, and the throughput-power relationship is written as:

$$H(P) = A_s P + B_s \tag{12}$$

For the Off mode $(s = 1)$, we define:

$$A_1 = 0, \qquad B_1 = 0 \tag{13}$$

So that $H(P) = 0$ when the unit is de-energized. For consistency with the operating-power constraints used later in Section IV, the feasible power range of the Off mode is also defined as $L_1 = 0$ and $U_1 = 0$.

For the active operating modes $s \in \{2, 3, 4\}$, the coefficients are given by:

$$\begin{aligned} A_s &= \frac{k_{thru}(1-p_{pool})}{\alpha C_{eff} V_{nom}^2} \left(\eta - (a_T + k_T)\kappa\right) \\ B_s &= -\frac{k_{thru}(1-p_{pool})}{\alpha C_{eff} V_{nom}^2} \left(P_{aux} + P_{fan,0} + P_{leak,0}\right) \end{aligned} \tag{14}$$

Accordingly, the throughput-power mapping is applied within the feasible range of each operating mode. This linear relationship $H(P)$ is the core achievement of this physical modeling, as it clearly quantifies the conversion efficiency from P to H.

B. Economic Revenue Calculation

Based on the mapping from P to H, this section constructs an economic model that converts computational throughput into revenue. For throughput-oriented or quantity-settled flexible computing services, the throughput-to-revenue relationship is represented by (15), and the corresponding interval revenue is $mH\Delta t$.

$$M = mH \tag{15}$$

Here, m denotes the revenue coefficient per unit effective computational throughput rate per unit time. In practical applications, m is obtained by converting workload-specific prices, rewards, fees, or internal settlement terms into a consistent economic value. For latency-sensitive or strict SLA-driven services, the economic layer should instead be replaced by a richer utility or penalty model.

This section uses the Bitcoin mining business as an example to specifically illustrate the composition of the unit computational throughput revenue m. This business revenue directly depends on the market price of the cryptocurrency, the entire network's mining difficulty, and the mining pool's settlement method. Based on the mainstream pooled operation model, a standardized unit computational throughput revenue model can be constructed. Assuming the FPPS (Full Pay Per Share) settlement method is adopted, meaning both protocol rewards

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TABLE I
COINCIDENT PEAK MECHANISMS

Market	Mechanism	Affected Charge Type	Settlement Period	Calculation Method
ERCOT [12]	Four Coincident Peaks	Transmission cost charge	June-September	Average power of the 15-minute peak interval each month
IESO [14]	Top-5 Peaks	Global adjustment related charges	May 1–April 30 base period	Electricity consumption during the 5 peak hours
PJM [32]	Five Coincident Peaks	Capacity charge	June–September	Average power of the 5 peak hours

and transaction fees are included at their expected values, m can be expressed as:

$$m = P_{asset}\frac{3600 \cdot 10^{12}}{D \cdot 2^{32}}\big(R^{\mathrm{blk}}+E[F]\big)(1-f_{pool})(1-s_l) \quad (16)$$

The factor 3600 aligns the second-based throughput unit with the hourly scheduling interval.

III. CP PREDICTION STRATEGY

The CP mechanism represents a significant component of FDC electricity costs. To provide critical inputs for the co-optimization model, this section first analyzes this mechanism, then constructs a formal CP prediction strategy. This strategy is designed to translate market rules and probabilistic forecasts into actionable decision inputs.

A. CP Mechanism

In many electricity markets, capacity costs for large consumers are allocated using a CP mechanism [12]. This mechanism can fairly distribute system capacity costs and incentivize users to actively participate in demand-side response, thereby enhancing the overall operational efficiency and stability of the grid [11].

The core of the CP mechanism is that a user's capacity charge is not determined by their own peak power usage, but by their average power usage during the peak periods when the entire grid (or a specific region) load reaches its highest peak. This average value, the peak load contribution, serves as the capacity cost baseline for the next settlement period [32]. Unlike time-of-use tariffs, which are priced in advance by time block, the key difficulty of CP lies in its ex-post determination. The true peak periods can only be confirmed after the complete settlement period has ended. This characteristic also leads to uncertainty in prediction and action [14].

This mechanism has different implementation forms in various global markets, but the principle is similar, as shown in Table I. For FDCs, the CP mechanism is both a huge cost-saving opportunity and a severe prediction challenge. Successful identification and peak shaving can bring significant annual savings, while misjudgment can lead to financial losses and potential business opportunity costs [21]. Therefore, the accurate prediction of CP events is the core challenge for FDCs to mitigate risks and save costs, and is also the starting point for constructing the prediction strategy in the subsequent part of this section.

Algorithm 1: CP Prediction Strategy

Input: Rules; D_{pred}; M_{day}; M_{hour}; θ.
Output: P_{day}; $P_{h|\mathrm{day}}$; Path.

$P_{\mathrm{day}} \leftarrow 0$;
$P_{h|\mathrm{day}} \leftarrow$ null ;
Path $\leftarrow$ LowRisk ;
if *CheckRules(D_{pred}, Rules)* = ***true*** **then**
  $P_{\mathrm{day}} \leftarrow M_{\mathrm{day}}$.predict$(D_{\mathrm{pred}})$;
  if $P_{\mathrm{day}} \geq \theta$ **then**
    $p_{\mathrm{raw}} \leftarrow M_{\mathrm{hour}}$.predict$(D_{\mathrm{pred}})$;
    $P_{h|\mathrm{day}} \leftarrow$ Normalize(p_{raw}) ;
    Path $\leftarrow$ HighRisk ;
  end
end
return $(P_{\mathrm{day}}, P_{h|\mathrm{day}}, \mathrm{Path})$;

B. General CP Prediction Strategy

This section constructs a CP prediction strategy as a structured interface between probabilistic forecasting and operational optimization. Unlike end-to-end decision-focused learning, the proposed methodology remains modular, with the prediction models trained separately and coupled to the operation model through explicit probabilistic outputs.

In this paper, the decision-relevant CP risk information consists of the day-level CP occurrence probability P_{day} for risk-path switching and the conditional hourly CP probability vector $P_{h|\mathrm{day}}$ for scenario construction and weighting. Other prediction outputs may still be useful for forecasting, but they are not directly used in the proposed adaptive co-optimization model.

To extract these quantities in a structured and operationally usable manner, the prediction strategy is organized into the following three stages:

1) Rule-based Filtering: This step performs an initial screening of all days based on the market's deterministic rules (e.g., settlement periods, holidays). This is different from the simple practice of using calendar information as one of the input features for a machine learning model [20]. This methodology solidifies domain knowledge into hard constraints through a deterministic filtering module and significantly narrows the analysis scope for subsequent probabilistic models.

2) Peak-Day Identification: For candidate days that pass the initial screening, various advanced prediction algorithms such as Feedforward Neural Network, Random Forest, Extreme Gradient Boosting, etc., are used to conduct risk assessment, identify high-risk days, and output their probability P_{day}. This step addresses the peak-day identification problem commonly focused on in the literature [14]–[16], and can flexibly embed any advanced classification algorithm as its core engine.

3) Peak-Hour Localization: Only when a day's probability P_{day} exceeds θ is this stage activated. It performs peak-hour prediction for this high-risk day and outputs a conditional

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probability vector $P_{h|\mathrm{day}}$. This conditional activation mechanism focuses computational resources on high-risk events, forming an efficient, cascading decision funnel.

The complete logic of this three-stage prediction strategy is formally described in Algorithm 1. Through this process, the methodology extracts the probabilistic inputs required by the adaptive co-optimization model, namely P_{day} for risk-path switching in Section IV.B and $P_{h|\mathrm{day}}$ for scenario construction in Section IV.A.

IV. Co-optimization Methodology

The day-ahead operation of an FDC is naturally formulated as a two-stage decision problem that requires balancing baseline schedules, AS bids, and the financial risk associated with CP uncertainty. To address this, this section constructs the co-optimization methodology to co-optimize all revenue streams and costs. The section details the mathematical formulation and demonstrates its integration into the risk-driven adaptive decision methodology.

A. Two-Stage Stochastic Programming Model

The FDC's operation on a high-risk CP day is formulated as a two-stage stochastic model. This model determines optimal first-stage (day-ahead) decisions while accounting for second-stage (real-time) operational adjustments in response to uncertainty. The objective is to maximize the operator's expected total profit.

*1) **Uncertainty Modeling and Decision Variables**:* The model solves the conditional optimization problem for a high-risk CP day. The primary uncertainty is the timing of the peak hour. We define Ω as the set of scenarios, where $\omega \in \Omega$ represents the CP event occurring at hour ω. The CP indicator $W_{t,\omega}$ is a pre-defined binary parameter derived from the prediction stage. For each scenario ω, $W_{t,\omega}$ is fixed as a constant coefficient, thereby ensuring that all constraints and the objective function maintain a linear structure relative to the decision variables. The scenario probability π_ω is given by $P_{h|\mathrm{day}}$ (from Section III) and satisfies:

$$\sum_{\omega\in\Omega} \pi_\omega = 1, \pi_\omega \quad \geq 0 \tag{17}$$

The model's decisions are structured in two stages. The first-stage decisions determine the baseline operation schedule, including Q_t^{DA} and $r_{k,t}$. The second-stage decisions are scenario-dependent ($\forall\omega \in \Omega$) and include $P_{t,\omega}^{eff}$ and the unit-level adjustments for AS deployment ($dw_{i,t,s}^{\omega}$, $uw_{i,t,s}^{\omega}$) and CP-avoidance curtailment ($gw_{i,t,s}^{\omega}$). The throughput quantities H_t^{base} and $H_{t,\omega}^{eff}$ are auxiliary variables introduced for compactness and economic interpretability; they are uniquely determined by the power decisions through the affine power-throughput mapping and do not add extra degrees of freedom to the optimization problem.

*2) **Objective Function**:* The objective is to maximize the expected total profit Π, defined as the sum of the deterministic first-stage profit Π_{DA} and the expected second-stage profit $\Pi_{RT}(\omega)$ over all scenarios:

$$\max \Pi = \Pi_{DA} + \sum_{\omega\in\Omega} \pi_\omega \Pi_{RT}(\omega) \tag{18}$$

While AS settlements may include mileage payments, deployment-related revenues, or non-compliance penalties, this paper models capacity payments only and does not explicitly represent performance-penalty settlements. This choice is made to isolate the main operational coupling studied here, namely the trade-off between flexibility reservation for AS and load curtailment for CP avoidance. Under this modeling scope, conflicting obligations are coordinated primarily through feasibility constraints rather than through ex-post penalty terms. Including additional settlement or penalty terms would change the absolute profitability results, but would not alter the core coordination mechanism developed in this paper.

Π_{DA} comprises the AS capacity revenue less the day-ahead energy cost:

$$\Pi_{DA} = \sum_t (\sum_k \alpha_{k,t} r_{k,t} - p_t^{DA} Q_t^{DA}) \Delta t \tag{19}$$

$\Pi_{RT}(\omega)$ accounts for the real-time computing revenue, the energy cost deviation settled in the real-time market, and the expected CP penalty. Consistent with Section II-B, the interval computing revenue is $mH_{t,\omega}^{eff}\Delta t$:

$$\begin{aligned}\Pi_{RT}(\omega) = \sum_t (mH_{t,\omega}^{eff} - p_t^{RT}(P_{t,\omega}^{eff} - Q_t^{DA}))\Delta t \\ - \sum_t C^{CP} W_{t,\omega} P_{t,\omega}^{eff} \Delta t\end{aligned} \tag{20}$$

*3) **First-Stage Constraints (Day-Ahead)**:* These constraints establish the day-ahead baseline operation (P_t^{base}, H_t^{base}) and ensure all AS capacity bids are physically feasible.

a) Equipment operating mode: Each unit i must select exactly one operating mode s at time t:

$$\sum_s y_{i,t,s} = 1, \ y_{i,t,s} \in \{0,1\}, \ \forall i,t,s \tag{21}$$

b) Computing unit operating power: The baseline power $p_{i,t,s}$ is constrained by the selected mode's limits (L_s, U_s):

$$L_s y_{i,t,s} \leq p_{i,t,s} \leq U_s y_{i,t,s} \tag{22}$$

c) Baseline aggregation: The total baseline power is aggregated from all units, and the corresponding baseline throughput follows from the affine mapping in (12):

$$P_t^{base} = \sum_{i,s} p_{i,t,s} \tag{23}$$

$$H_t^{base} = \sum_{i,s} (A_s p_{i,t,s} + B_s y_{i,t,s}) \tag{24}$$

d) Facility power limits: The total accessible power range $(P_{\min,t}, P_{\max,t})$ is determined by the sum of active mode limits:

$$P_{\min,t} = \sum_{i,s} L_s y_{i,t,s}, \ P_{\max,t} = \sum_{i,s} U_s y_{i,t,s} \tag{25}$$

Q_t^{DA} is bounded by the baseline:

$$0 \leq Q_t^{DA} \leq P_t^{base} \tag{26}$$

e) Ancillary service capacity: Offered $r_{k,t}$ for downward (AS_D) and upward (AS_U) services is constrained by the

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facility's available headroom and foot-room relative to P_t^{base}:

$$\sum_{k\in AS_D} r_{k,t} \le P_t^{base} - P_{\min,t} \tag{27}$$

$$\sum_{k\in AS_U} r_{k,t} \le P_{\max,t} - P_t^{base} \tag{28}$$

f) Thermal Shock Prevention: Rapid power fluctuations cause drastic temperature changes that induce thermal stress and shorten chip lifespan. Thus, direct transitions between the Off mode and the High mode are prohibited:

$$y_{i,t,4} + y_{i,t+1,1} \le 1, \quad \forall i, t \in \{1, \dots, |T|-1\} \tag{29}$$

$$y_{i,t,1} + y_{i,t+1,4} \le 1, \quad \forall i, t \in \{1, \dots, |T|-1\} \tag{30}$$

*4) **Second-Stage Constraints (Real-Time)**:* These constraints are enforced for each scenario $\omega \in \Omega$, ensuring that all real-time adjustments adhere to the physical limits and the first-stage AS commitments.

a) AS deployment limits: Real-time AS deployments $(dw_{i,t,s}^{\omega}, uw_{i,t,s}^{\omega})$ on each unit are bounded by the available headroom/foot-room within its selected mode:

$$\begin{aligned} 0 &\le dw_{i,t,s}^{\omega} \le p_{i,t,s} - L_s y_{i,t,s} \\ 0 &\le uw_{i,t,s}^{\omega} \le U_s y_{i,t,s} - p_{i,t,s} \end{aligned} \tag{31}$$

b) CP curtailment: Additional curtailment $gw_{i,t,s}^{\omega}$ is subordinate to AS deployment:

$$0 \le gw_{i,t,s}^{\omega} \le W_{t,\omega}(p_{i,t,s} - L_s y_{i,t,s} - dw_{i,t,s}^{\omega}), \forall i, t, s, \omega \tag{32}$$

It is permitted only during the CP hour ($W_{t,\omega} = 1$) and is limited by the residual downward flexibility after AS deployment. Therefore, in the modeled setting, CP avoidance cannot consume the portion of flexibility already used for AS delivery, which provides the main coordination mechanism between the two market obligations.

c) Deployment aggregation and capacity: Aggregated deployments $(dP_t^{\omega}, uP_t^{\omega})$ must respect the total AS capacity $r_{k,t}$ bid in the first stage and the total facility limits:

$$dP_t^{\omega} = \sum_{i,s} dw_{i,t,s}^{\omega} \tag{33}$$

$$uP_t^{\omega} = \sum_{i,s} uw_{i,t,s}^{\omega} \tag{34}$$

$$gP_t^{\omega} = \sum_{i,s} gw_{i,t,s}^{\omega} \tag{35}$$

$$dP_t^{\omega} \le \sum_{k\in AS_D} r_{k,t} \tag{36}$$

$$uP_t^{\omega} \le \sum_{k\in AS_U} r_{k,t} \tag{37}$$

$$\begin{aligned} dP_t^{\omega} &\le P_t^{base} - P_{\min,t}, \\ P_t^{base} + uP_t^{\omega} &\le P_{\max,t} \end{aligned} \tag{38}$$

d) Effective power and throughput: The final real-time power is obtained by applying all adjustments to the baseline values, and the corresponding throughput follows from the same affine mapping:

$$P_{t,\omega}^{eff} = P_t^{base} - dP_t^{\omega} - gP_t^{\omega} + uP_t^{\omega} \tag{39}$$

$$H_{t,\omega}^{eff} = H_t^{base} + \sum_{i,s} A_s(uw_{i,t,s}^{\omega} - dw_{i,t,s}^{\omega} - gw_{i,t,s}^{\omega}) \tag{40}$$

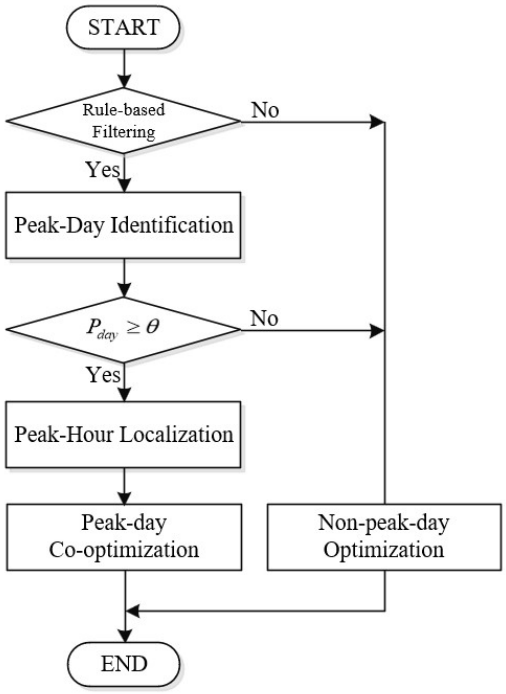


Fig. 1. Risk-Driven Adaptive Optimization Methodology

B. Risk-Driven Adaptive Optimization Methodology

The proposed model in Section IV.A is conditional, as a single unified stochastic model would be computationally intractable. This section presents the RPCM, which integrates the paper's core components into a single cohesive framework. The primary effect of this methodology is to provide a robust decision path that intelligently balances decision optimality with computational efficiency. It achieves this by using the predicted CP risk as a dynamic switch. This adaptive structure, illustrated in Fig. 1, ensures the intensive stochastic optimization is reserved only for days where financially justified by the predicted risk. The six steps below formalize this logic, beginning with a one-time foundational setup in Step 1, followed by the five-stage daily decision flow in Steps 2–6.

1) Step 1 (FDC Model Initialization): This foundational step defines the FDC's characteristics. Based on the selected FDC type (e.g., Bitcoin mining, HPC), the high-fidelity physical and economic models from Section II are established, quantifying the power-to-throughput (P-H) relationship via (12) and the throughput-to-revenue (H-M) relationship via (15). This provides the essential parameters (A_s, B_s, m) for the optimization.

2) Step 2 (Deterministic Rule-based Filtering): Upon receiving day-ahead information, this step screens the target

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day against deterministic rules (i.e., within the CP settlement window, not a weekend or holiday). If the day fails this check, it is assigned zero CP risk ($P_{\text{day}} = 0$) and the process moves to Step 6. Otherwise, it is marked as a potential CP candidate and advances to Step 3.

3) Step 3 (Peak-Day Identification and Risk Classification): Potential CP candidates are assessed using the peak-day model M_{day} to obtain the probability P_{day}. This probability is then compared to the risk threshold θ. If $P_{\text{day}} \geq \theta$, the day is classified as high-risk and proceeds to Step 4. Otherwise, it is deemed low-risk and moves to Step 6.

4) Step 4 (Peak-Hour Localization): For high-risk days, the peak-hour model M_{hour} performs a refined hour-by-hour probabilistic assessment. It outputs the conditional probability vector $P_{h|\text{day}}$, which serves as the input for the co-optimization stage in Step 5.

5) Step 5 (CP Co-optimization, High-Risk Branch): The full stochastic model in Section IV.A is activated, using the $P_{h|\text{day}}$ vector from Step 4 as the scenario probabilities π_ω. The objective function (18)–(20) is then solved to co-optimize revenue while minimizing the expected CP penalty cost.

6) Step 6 (Conventional Profit Maximization, Low/No-Risk Branch): Triggered by either Step 2 or Step 3, this low/no-risk branch activates a conventional deterministic model. The optimization objective degenerates to maximizing operational profit by setting the CP penalty term C^{CP} in (20) to zero, focusing only on the balance of computing revenue, AS revenue, and energy costs.

Overall, the RPCM preserves CP-aware decision robustness on predicted high-risk days while avoiding unnecessary stochastic optimization on low/no-risk days. Because predicted CP risk only determines branch activation, false positives may lead to economically conservative operation but do not directly enforce curtailment; the operating schedule remains endogenously determined by the co-optimization model, and the threshold θ mitigates over-sensitive switching. Conversely, false negatives may route a true high-risk day to the low/no-risk branch and thereby expose the operator to realized CP charges. In practice, the threshold θ can be selected conservatively to reduce missed high-risk days, at the expense of more frequent activation of the stochastic branch.

C. Discussion on Model Portability

The proposed RPCM is designed with a modular architecture to improve portability across different electricity markets. The physical-layer mapping and the risk-driven decision logic are general components, whereas the market-layer implementation is market-specific. In particular, when the methodology is transferred to another regulatory environment, the operator must update the CP settlement rule, the penalty parameter C^{CP}, the AS product definitions k, the localized prediction feature set D_{pred}, and the revenue parameter m associated with the business model of the flexible data center. In markets with different capacity-charge designs, such as top-N peak settlement or multi-interval settlement rules, the market-layer settlement equations and scenario construction should be reformulated accordingly. This also implies corresponding updates to the scenario set Ω, the indicator parameter $W_{t,\omega}$, and the scenario probability π_ω. By contrast, the underlying power-throughput mapping and the adaptive prediction-to-decision architecture remain unchanged.

TABLE II
ANTMINER S17-PRO-53T OPERATING PARAMETERS

Mode	$H_{\min}$ [TH/s]	$H_{\max}$ [TH/s]	$P_{\min}$ [W]	$P_{\max}$ [W]
Off	0	0	0	0
Low	36	48	1296	2079
Normal	53	55.95	2094	2568
High	50	62	2250	3371

TABLE III
BITCOIN REVENUE PARAMETERS

Symbol	Value	Symbol	Value
P_{asset}	\$101,000/BTC	D	130×10^{12}
R^{blk}	3.125 BTC/block	$E[F]$	0.2 BTC/block
f_{pool}	0.01	s_l	0.005

V. CASE STUDY

This section validates the effectiveness of the RPCM through a case study analysis. The case study environment is set in the ERCOT market in Texas, with a Bitcoin mining data center as the case study subject. The aim is to validate that the proposed methodology can effectively identify CP risks and execute optimal avoidance strategies, and to demonstrate the economic value brought by the co-optimization model compared to isolated decisions.

A. Parameter Setting

Bitcoin mining is adopted here as a representative high-flexibility, throughput-oriented workload, for which the service-side revenue is closely tied to delivered computational throughput and the workload exhibits negligible job dependency and no strict task-level completion deadlines. This setting provides a clean benchmark for isolating the core coordination mechanism between CP management and AS participation.

1) FDC Model Initialization: This case study considers a 10-unit FDC composed of commercial ASIC miners (Antminer S17 Pro-53T [33]). The three active operating modes listed in Table II define the piecewise linear P–H mapping in (41), while the Off mode is separately defined by $P = 0$ and $H = 0$.

$$H(P) = \begin{cases} 0.01533\,P + 16.14, & 1296 \leq P \leq 2079 \\ 0.006224\,P + 39.97, & 2094 \leq P \leq 2568 \\ 0.010705\,P + 25.91, & 2250 \leq P \leq 3371 \end{cases} \quad (41)$$

Using (16) and the parameters in Table III, the marginal revenue coefficient is calculated as $m = \$2.13 \times 10^{-3}/(\text{TH/s}\cdot h)$, consistent with the hourly scheduling interval used in the case study.

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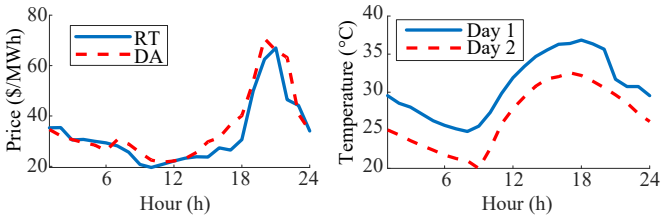


Fig. 2. ERCOT Energy Prices (left) and Ambient Temperatures for Day 1 and Day 2 (right)

TABLE IV
INPUT FEATURES FOR PREDICTION MODELS

Category	Features
Time	Hour index; weekend/holiday indicators
Air Temp.	Hourly temperature (current, $t \pm 3$h); daily min/max temperature; 7-day-ahead min/max temperature
Dew Point	Hourly dew point (current, $t \pm 3$h)
Load	Month-to-date peak load; previous-day peak load; previous-day load (same hour)

2) Market and Typical Days: We set the market environment to the ERCOT day-ahead (DA) and real-time (RT) markets [34], with a time resolution of 1 hour. The electricity prices are shown in Fig. 2 (left). We consider five types of ASs: Regulation Up Service (RU), Regulation Down Service (RD), Responsive Reserve Service (RRS), ERCOT Contingency Reserve Service (ECRS), and Non-Spinning Reserve (Non-Spin). We calculate CP charges based on the average power during the 1-hour system peak period of each month from June to September. We set C^{CP} to \$15,000/MWh. Because the optimization model uses a 1-hour time resolution whereas the ERCOT settlement is based on a 15-minute peak interval, C^{CP} is treated here as an equivalent hourly CP penalty coefficient obtained from the original settlement rule. To compare strategies under different risks, we select three typical days:

1) Day 1: A high-temperature day (2025-08-18)
2) Day 2: A normal day (2025-09-17)
3) Day 3: A non-CP window day (2025-05-08)

The temperatures for Day 1 and Day 2 are shown in Fig. 2 (right).

3) Prediction Model Parameters: We use a Random Forest classifier for both peak-day identification and peak-hour localization. The main input features are summarized in Table IV, while the key prediction settings are summarized in Table V. The prediction module follows a chronological out-of-sample day-ahead forecasting protocol. Electricity prices are excluded from the CP prediction feature set and are used only in the subsequent optimization stage. The risk threshold θ is set to 3%. If $P_{\text{day}} \geq \theta$, the methodology activates the stochastic model (high-risk branch); otherwise, the conventional deterministic model (low/no-risk branch) is used.

TABLE V
PREDICTION MODULE SETTINGS

Aspect	Specification
Data source and range	ERA5 and ERCOT load websites [35], [36]; hourly load, temperature, and dew-point data from Jan. 2015 to Dec. 2024
Forecast setting	Chronological out-of-sample day-ahead forecasting
Preprocessing	Sorting, deduplication, hourly alignment by linear interpolation; missing-load samples removed; remaining missing features filled with zeros
Rule-based screening	June–September CP months and calendar conditions
Peak-day model	Random Forest (400 trees); label: monthly Top-1 peak day
Peak-hour model	Random Forest (400 trees); label: hour of daily maximum load
Training scheme	Peak-day model trained on June–September samples; peak-hour model uses month-specific June–September models with pooled fallback
Non-CP-month output	Peak-day probability set to 0; peak-hour probability set to a uniform 24-hour distribution
Excluded variable	Electricity prices are excluded from prediction and used only in optimization

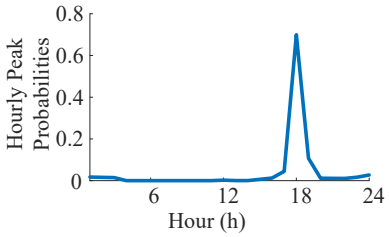


Fig. 3. Predicted Hourly CP Probabilities for Day 1 (High-Temperature Day)

B. Validation of the Risk-Driven Adaptive Decision Methodology

This section aims to validate that the proposed methodology can adaptively adjust operation strategies based on CP risk prediction results to achieve a balance between risk and revenue.

1) CP Risk Prediction and Decision Path Activation: The prediction model first performs deterministic rule-based filtering on different dates. Day 1 and Day 2 pass the filtering. Day 3 does not pass because it is not in the CP window period. Next, peak-day prediction is performed for Day 1 and Day 2. P_{day} for Day 2 is almost 0%, and P_{day} for Day 1 is 15.7%. Given $\theta = 3\%$, Day 2 is identified as a low-risk day, entering the conventional profit maximization decision path. Day 1 is identified as a high-risk day, successfully activating the peak-hour localization and CP co-optimization decision paths. Fig. 3 shows $P_{h|\text{day}}$ for Day 1. The peak-hour risk is mainly concentrated from 17:00 to 19:00, and reaches a peak of 69.96% at 18:00, which matches the high temperature at that time.

2) Comparison of Optimal Operation Strategies under Different Risks: As shown in Fig. 4, on the low-risk Day 2, the model's objective is conventional profit maximization. Its strategy involves fine-grained hourly arbitrage: switching to Mode 4 at 9:00 to maximize computing revenue, dropping to Mode 2 during the high-price period (19:00-23:00) to save

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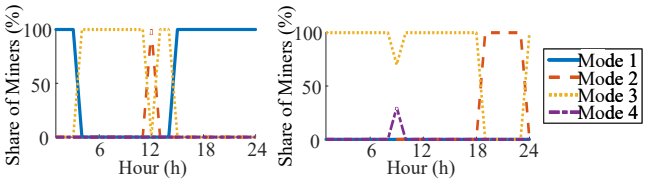


Fig. 4. Optimal Miner Operating Mode Distribution for Day 1 (High-Risk Day, Left) and Day 2 (Low-Risk Day, Right).

TABLE VI
BREAKDOWN BY TYPICAL DAY

	Hash[$]	AS[$]	DA[$]	RT[$]	CP[$]	Total[$]
Day 1	12.32	0.18	2.51	3.06	0.80	6.13
Day 2	25.38	1.85	5.00	9.15	—	13.08

costs, and maintaining Mode 3 otherwise. In sharp contrast, on the high-risk Day 1, the objective shifts to maximizing expected profit by avoiding costly CP penalties. This triggers an extreme risk-avoidance strategy: as $P_{h|\text{day}}$ increases, all miners are shut down from 15:00 to 24:00. They are also shut down during the 0:00-4:00 period, when the risk is slightly elevated. Consequently, the total output drops to zero during these risk periods.

This comparison demonstrates that the methodology successfully integrates prediction with optimization. Under realistic parameters where CP penalties far exceed the marginal computing revenue, the model ceases to be a simple price-follower. It transforms into a rational risk avoider, correctly judging that sacrificing short-term revenue (opportunity cost) is far more economical than bearing the potential, high-value CP capacity charges (risk cost).

3) Economic Benefit Analysis of Coordinated Avoidance: Table VI further quantifies the economics of this balance. In Tables VI–VIII, DA, RT, and CP denote cost components, while AS and Hash denote revenue components. The total profit for Day 1 is $6.13, which is lower than the total profit of $13.08 for Day 2. The reason for the sharp decline in profit is that an aggressive active peak-shaving strategy was executed to avoid high CP penalties, leading to the computing revenue (Hash) on Day 1 being only $12.32, far lower than the $25.38 on Day 2. The key is that this sacrifice is valuable. By actively shutting down during peak periods, the Day 1 strategy successfully controls the potential huge CP charges to an extremely low expected level, only $0.8. If the high-output strategy of Day 2 were adopted on Day 1, its CP penalty would be far higher than this. This proves the effectiveness of the proposed co-optimization methodology in quantifying the balance between computing revenue loss and CP risk avoidance.

C. Economic Value Analysis of the Co-optimization Model

This section quantifies the true economic value of the co-optimization methodology by analyzing two key typical scenarios, high-risk days (Day 1) and low/no-risk days (Day 2 and Day 3), which account for the vast majority of the year.

TABLE VII
STRATEGY COMPARISON FOR HIGH-RISK DAY

	Hash[$]	AS[$]	DA[$]	RT[$]	CP[$]	Total[$]
Strategy 1	25.28	1.83	5.00	9.00	294.05	-280.93
Strategy 2	12.32	0	2.24	3.35	0.80	5.93
Strategy 3	12.32	0.18	2.51	3.06	0.80	6.13

TABLE VIII
STRATEGY COMPARISON FOR LOW/NO-RISK DAYS

	Hash[$]	AS[$]	DA[$]	RT[$]	Total[$]
Strategy 4	24.12	—	—	12.95	11.17
Strategy 5	24.12	—	4.48	8.19	11.45
Strategy 6	25.38	1.85	5.00	9.15	13.08

It assesses the advantages of this methodology over isolated or simplified strategies.

1) Coordinated Avoidance Value in Day 1: This section compares three key strategies on this typical day: Strategy 1 (AS-oriented, ignoring CP risk), Strategy 2 (CP-oriented, not participating in AS), and Strategy 3 (the co-optimization model). As shown in Table VII, the comparison results clearly reveal the necessity of coordination. Although Strategy 1 obtained the highest computing revenue of $25.28 and AS revenue of $1.83, it failed to curtail load during high-risk periods. This resulted in a catastrophic $294.05 CP penalty, making its total profit $-280.93, the lowest of the three. This proves that an isolated pursuit of AS revenue while ignoring CP risk is infeasible. Strategy 2 successfully controlled the CP cost to $0.80, but it completely missed the $0.18 in AS capacity revenue, leading to a total profit of only $5.93. In contrast, Strategy 3 demonstrated superior coordinated value. It not only obtained $0.18 in AS revenue through optimization but also achieved the same level of $0.8 CP cost control, ultimately realizing the highest total profit of $6.13.

2) Multi-Market Participation Value in Low/No-Risk Days: Second, to validate the economic value on days with negligible CP risk, this section analyzes the strategy applied to both Day 2 and Day 3, which the methodology classifies as low/no-risk. This section compares three progressive decision strategies shown in Table VIII, using Day 3 data which yields the same profit structure as Day 2. The strategies are Strategy 4 (real-time market only), Strategy 5 (day-ahead + real-time market), and Strategy 6 (the co-optimization model). Strategy 4 is completely exposed to market volatility, leading to high electricity costs of $12.95. Strategy 5 effectively hedges against price volatility risk by locking in some low-priced electricity in the day-ahead market, increasing the total profit from $11.17 to $11.45. After introducing ASs, Strategy 6 obtains an additional revenue of $1.85 by selling AS capacity, further increasing the total profit to $13.08. This result indicates that on all days classified as low-risk, including non-CP window days, adopting the complete co-optimization strategy, compared to the baseline strategy of participating only in the real-time market, can bring a significant profit increase of up to 17.1%.

In summary, whether on high-risk CP days or on regular low/no-risk days, the co-optimization methodology proposed

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TABLE IX
SENSITIVITY OF TOTAL PROFIT TO EQUIVALENT CP PENALTY COEFFICIENT ON DAY 1

$/MWh	**3,750**	**7,500**	**15,000**	**30,000**	**60,000**
Strategy 1	-60.39	-133.90	-280.93	-574.97	-1163.06
Strategy 2	6.78	6.33	5.93	5.52	5.07
Strategy 3	7.19	6.53	6.13	5.73	5.29

in this paper can achieve significant economic gains compared to isolated or simplified strategies through refined multi-market balances.

3) Sensitivity to Equivalent CP Penalty Coefficient: The equivalent CP penalty coefficient C^{CP} depends on the settlement design, peak-interval definition, and time-resolution conversion adopted in a given market. To examine whether the conclusions of the case study are robust to this calibration, we perform a sensitivity analysis for the high-risk Day 1 by varying C^{CP} over a wide range, namely \$3,750/MWh, \$7,500/MWh, \$15,000/MWh, \$30,000/MWh, and \$60,000/MWh, corresponding to 0.25×, 0.5×, 1×, 2×, and 4× the benchmark value used in the main study.

As shown in Table IX, the profit of Strategy 1 (AS-oriented, ignoring CP risk) deteriorates rapidly as C^{CP} increases, indicating that this isolated strategy is highly vulnerable to the calibration of the equivalent CP penalty coefficient. By contrast, Strategy 2 (CP-oriented, without AS participation) and Strategy 3 (the proposed co-optimization model) remain much more stable across the tested range. More importantly, Strategy 3 consistently achieves the highest total profit for all tested values of C^{CP}.

These results show that although the absolute profit levels are affected by the equivalent penalty calibration, the relative strategy ranking remains unchanged. Therefore, the main conclusion of this paper is robust: explicitly coordinating CP avoidance and AS participation is consistently more effective than treating the two decisions separately, even when the equivalent CP penalty coefficient varies substantially.

D. Computational Performance

All simulations were performed on a workstation with 11th Gen Intel(R) Core(TM) i5-1135G7 @ 2.40GHz and 16 GB of RAM. The models were implemented in MATLAB using the Gurobi Optimizer version 11.0.3.

The computational results validate the efficiency of the adaptive methodology. The high-risk Day 1, which triggered the full 24-scenario stochastic model, solved in 44.84 seconds. In contrast, the low-risk Day 2 and non-CP Day 3, for which the methodology reverted to the simpler deterministic model, were solved in only 8.43 seconds and 2.29 seconds, respectively. This demonstrates that the methodology successfully balances decision optimality with computational tractability, reserving the intensive stochastic formulation only for days where high financial risk is identified.

While the case study uses a 10-unit cluster for demonstration, the formulation can be reformulated to improve tractability for large-scale facilities. For a homogeneous or grouped semi-homogeneous cluster, the unit-level binary variables can be replaced by aggregate integer variables that represent the number of units operating in each mode. The corresponding baseline power and real-time adjustment variables can also be aggregated at the mode level. Under this reformulation, the model size depends mainly on the number of modes, time periods, and scenarios, rather than directly on the total server count. For very large clusters, these aggregate integer variables may be further relaxed to continuous variables to obtain a linear-programming approximation, at the cost of introducing approximation error. The detailed development and validation of these scalable extensions will be addressed in future work.

VI. CONCLUSION

FDCs are increasingly positioned as grid-interactive resources that must simultaneously monetize their computational throughput, capture AS revenues, and avoid severe CP penalties. This paper develops a unified decision methodology to support these coupled operational goals for FDCs in CP-based electricity markets. Based on refined energy-flow analysis, we first establish a high-fidelity physical–economic model that links grid-side input power to effective computational throughput and then to revenue, yielding physically interpretable marginal costs and capacity limits for heterogeneous computing hardware. On top of this model, we design a three-stage CP risk prediction strategy that combines deterministic rule-based filtering with peak-day and peak-hour probabilistic assessments, producing day-level CP risk and hour-level conditional probabilities that are explicitly aligned with the requirements of the optimization model. This three-stage strategy is designed to be model-agnostic, focusing on the integration of probabilistic risk rather than a specific prediction algorithm. These components are integrated into a risk-driven and prediction-aware co-optimization methodology centered on a stochastic model, which co-optimizes AS bidding and baseline operation schedules while dynamically switching between CP-aware and conventional operation modes according to the predicted CP risk level.

A case study of a Bitcoin-mining FDC in the ERCOT market in Texas demonstrates the economic value of the proposed methodology. On high-risk CP days, the co-optimization strategy reduces the expected CP charge from a catastrophic \$294.05 under an AS-oriented strategy to only \$0.80, while still retaining AS revenue and achieving the highest total profit among all compared strategies. On low/no-risk days, jointly optimizing participation in the day-ahead, real-time, and AS markets increases total profit from \$11.17 under a real-time-only baseline to \$13.08, corresponding to a profit improvement of up to 17.1%. These results show that the methodology can endogenously balance CP risk avoidance against market opportunities, transforming FDCs from passive price-followers into rational risk-aware decision makers. The proposed physical–economic model and risk-driven prediction–optimization architecture are directly applicable to throughput-oriented and quantity-settled flexible computing loads participating in CP-based and multi-market environments.

Future work will extend the methodology in two directions. First, the service and economic layers will be generalized

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to more complex workloads, including latency-sensitive and SLA-driven services with job dependencies, queueing effects, and strict completion deadlines, by incorporating richer utility and penalty formulations, backlog dynamics, precedence constraints, and deadline-related constraints. Second, the case-study scope will be expanded from a single-facility setting to seasonal and fleet-level scenarios, together with a more detailed validation of CP prediction performance, threshold selection, and participation in other grid services and market designs, while preserving the same CP-aware prediction-to-decision architecture.

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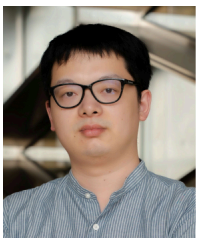
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